LFP Cell Single Particle Modeling

Zahid Sarıgöl

The target cell is 2.5 Ah Lithium Iron Phosphate (LFP), the parameters used in Battery Single Particle are given below:

Anode thickness	3.4E-5	m
Electrolyte layer count of anode	10	
Separator thickness	2.5E-5	m
Electrolyte layer count of separator	5	
Cathode thickness	8E-5	m
Electrolyte layer count of cathode	10	
Electrode plate area	1.738E-1	m^2
Anode particle radius	3.5E-6	m
Anode shell count	10	
Cathode particle radius	5.2E-8	m
Cathode shell count	10	
Electrodes Properties		
Volume fraction of anode active material	0.56	
Volume fraction of cathode active material	0.454	
Stoichiometry breakpoints specification	Normalized	
Normalized stoichiometry breakpoints	linspace(0.025, 0.975, 39)	
Anode open-circuit potential relative to n...	[.484; .335; .253; .209; .2; .181; .165;...	V
Cathode open-circuit potential relative to ...	[3.81;3.608;3.512;3.478;3.459;3.453;...	V
Anode maximum ion concentration	31900	mol/m^3
Anode maximum stoichiometry	0.811	
Anode minimum stoichiometry	0.0132	
Cathode maximum ion concentration	20950	mol/m^3
Cathode maximum stoichiometry	0.74	
Cathode minimum stoichiometry	0.035	
Diffusion coefficient of anode active mate...	3E-15	m^2/s
Diffusion coefficient of cathode active mat...	8E-18	m^2/s
Anode conductivity	200	S/m
Cathode conductivity	0.1	S/m

The parameters highlighted in yellow have been updated from their default values based on information from the literature [1], [2], [3].

Geometry	Electrode plate area
Geometry	Anode particle radius
Geometry	Anode shell count
Geometry	Cathode particle radius
Geometry	Cathode shell count
Electrodes Properties	Volume fraction of anode active material
Electrodes Properties	Volume fraction of cathode active material
Electrodes Properties	Stoichiometry breakpoints specification
Electrodes Properties	Normalized stoichiometry breakpoints
Electrodes Properties	Anode open-circuit potential relative to n...
Electrodes Properties	Cathode open-circuit potential relative to ...
Electrodes Properties	Anode maximum ion concentration
Electrodes Properties	Anode maximum stoichiometry
Electrodes Properties	Anode minimum stoichiometry
Electrodes Properties	Cathode maximum ion concentration
Electrodes Properties	Cathode maximum stoichiometry
Electrodes Properties	Cathode minimum stoichiometry
Electrodes Properties	Diffusion coefficient of anode active mat...
Electrodes Properties	Diffusion coefficient of cathode active mat...
Electrodes Properties	Anode conductivity
Electrodes Properties	Cathode conductivity

The <u>Electrode Plate Area</u> is calculated using the tuned parameters to model 2.5 Ah LFP cell:

$$Q = (A * Lc * \varepsilon c * c_{max,c} * (\theta_{max} - \theta_{min}) * F) / 3600$$

Q = Theoretical Capacity (2.5 Ah)
A = Electrode Plate Area (m^2)
Lc = Cathode thickness ($8 * 10^{-5}$ m)
εc = Active material volume fraction of cathode (0.454)
$c_{max,c}$ = Maximum ion concentration of LFP (20950 mol/m^3)
$(\theta_{max} - \theta_{min})$ = Stoichiometry range
F = Faraday constant (96485 C/mol)
3600 = Seconds to hours

$$2.5 = (A * (8 * 10^{-5}) * 0.454 * 20950 * (0.74 - 0.035) * 96485) / 3600$$

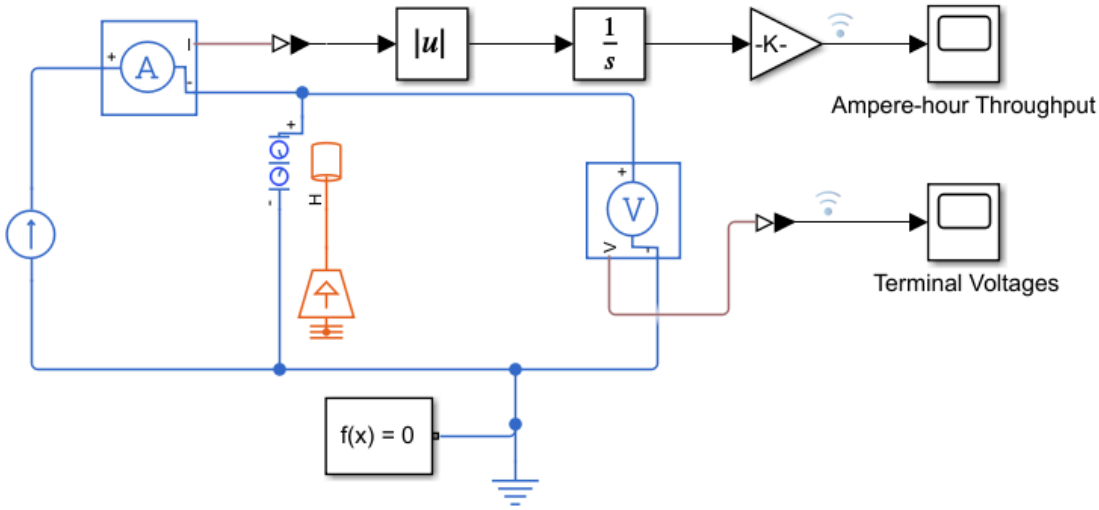
$$\underline{A = 0.17388\ m^2}$$

The experiment required to get the diffusion coefficient and the resistance is EIS (Electrochemical Impedance Spectroscopy), GITT (Galvanostatic Intermittent Titration Technique) can give the OCV and the diffusion coefficient as well. The particle sizes can be obtained using Scanning electron microscopy (SEM).

DC Current Source and Stop Time:

1C: 2.5, - 2.5 A.	Time: 3600 s
C/30: 0.083, - 0.083 A.	Time: (3600 * 30) s

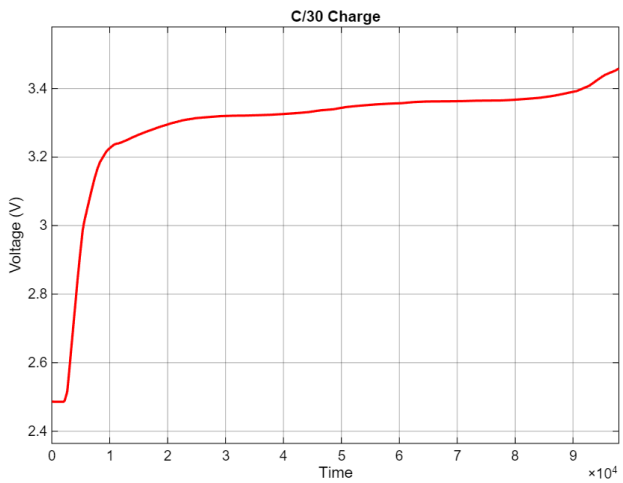
Single Particle Model (SPM) Setup:

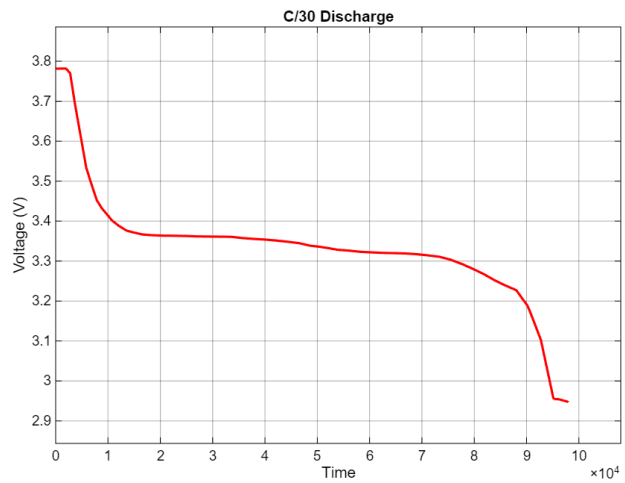


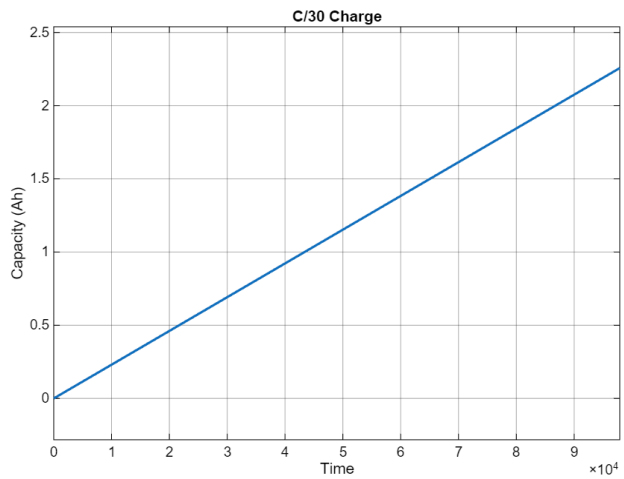
Battery Single Particle:	2.5 Ah LFP Battery
DC Current Source:	1C (2.5 A) and C/30 (2.5/30 = 0.083 A)
Solver Configuration (f(x) = 0):	Differential equations
Electrical Reference:	Ground
Thermal Reference:	25 C
Voltage Sensor:	Terminal voltage
Current Sensor:	To get the capacity
PS-Simulink Converters:	Simscape signals to mathematical signals
Absolute Value:	To always get positive capacity
Integrator (1/s):	Current over time
Gain (-K-):	* 1/3600 (to get hour)
Scopes:	Plots

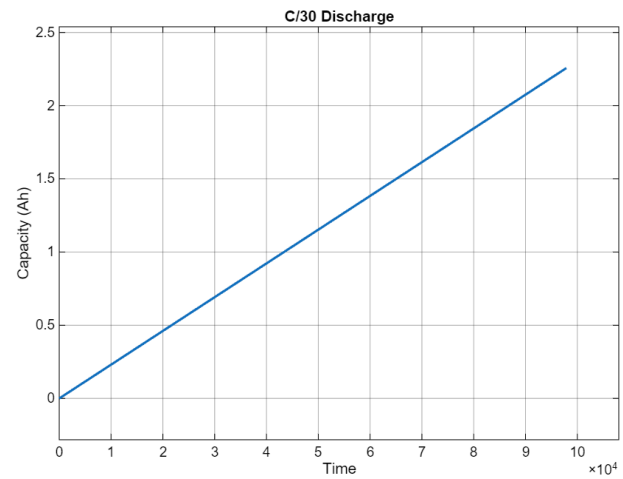
The model was run by changing the current value in DC Current Source (+-0.083 for C/30, +-2.5 for 1C), Stop Time (3600*30 s for C/30, 3600 s for 1C) and the Initial Targets SOC in the Battery Single Particle (0 for charging, 1 for discharging).

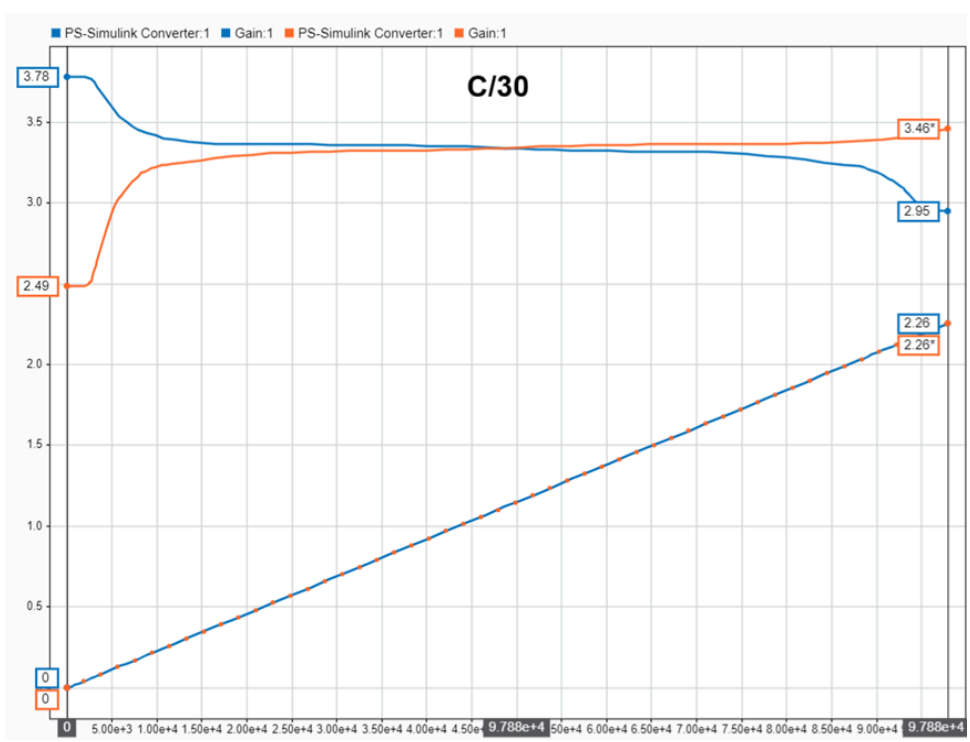
Plots: **C/30**


C/30 Charge
Voltage (V)
Time
×10^4

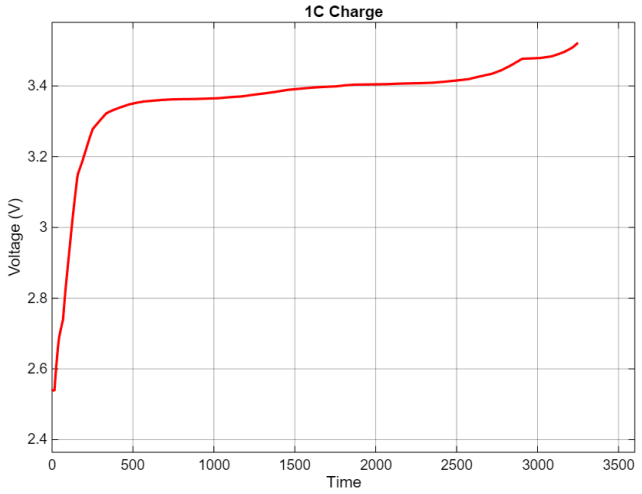

C/30 Discharge
Voltage (V)
Time
×10^4

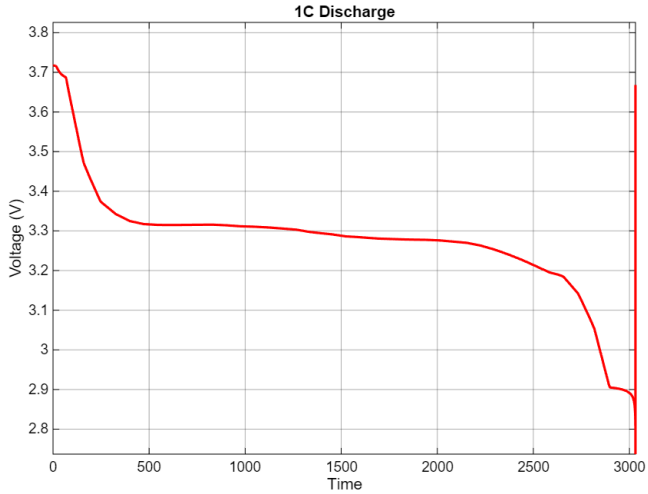

C/30 Charge
Capacity (Ah)
Time
×10^4

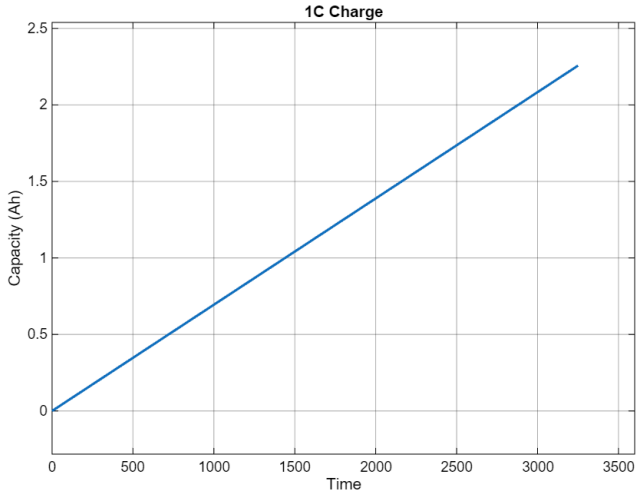

C/30 Discharge
Capacity (Ah)
Time
×10^4

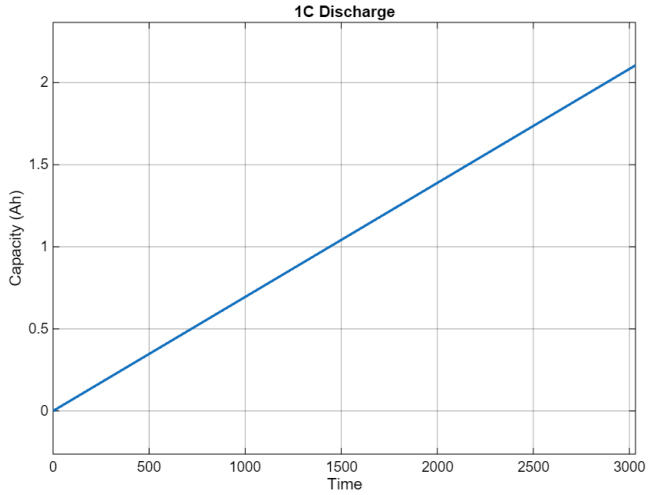

PS-Simulink Converter:1
Gain:1
PS-Simulink Converter:1
Gain:1
C/30
3.78
3.46*
2.95
2.49
2.26
2.26*
0
0

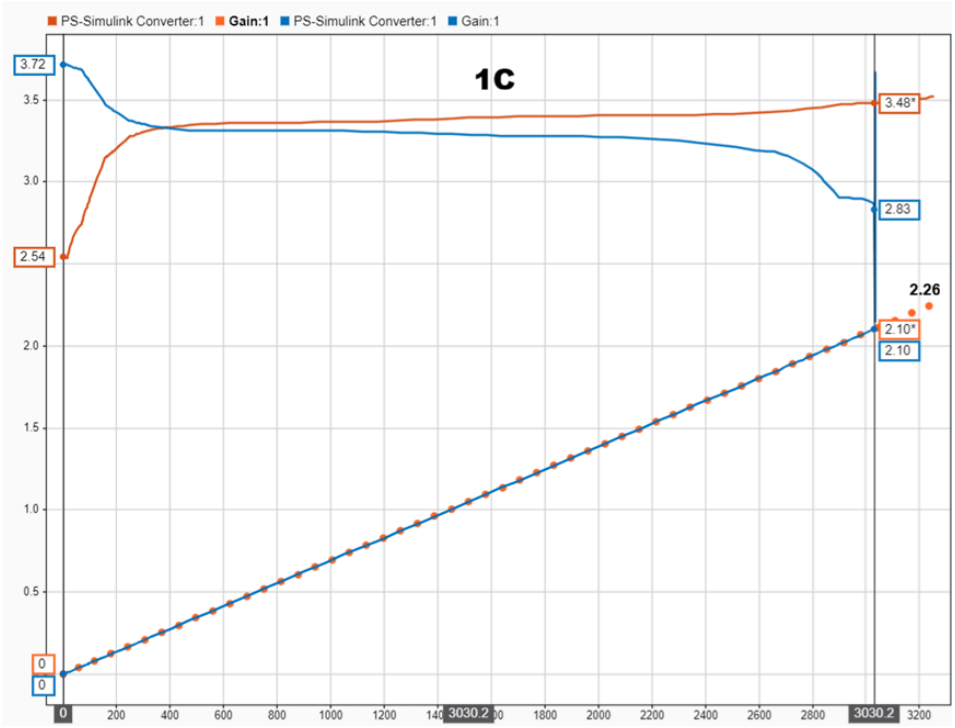
1C


1C Charge
Voltage (V)
Time


1C Discharge
Voltage (V)
Time


1C Charge
Capacity (Ah)
Time


1C Discharge
Capacity (Ah)
Time


PS-Simulink Converter:1
Gain:1
PS-Simulink Converter:1
Gain:1
1C
3.72
3.48*
2.83
2.54
2.26
2.10*
2.10
0
0

The flat plateau around 3.3 to 3.4 V shows the LFP's characteristic behaviour. The theoretical capacity was designed to be 2.5Ah, but the model's discharge capacity is around 2.26 Ah for C/30 and 2.10 Ah for 1C rates due to the practical limitations like diffusion resistance. As expected, the capacity at slow discharge rate C/30 is higher than the fast 1C rate because of the diffusion limits.

Discussion:

The parameters were taken from the reference literatures, a capacity calculation was performed to get the target capacity of 2.5 Ah. Using the theoretical capacity formula, the required electrode area was calculated. The SPM circuit was arranged to get the Terminal voltage and the Ampere-hour throughput.
SPM successfully showed the characteristic voltage profile of LFP. As expected, in all cases the capacity was lower than the theoretical capacity, and even lower when the discharge rate was higher.
The results show the usable capacity is dependent on the applied current rate, and is consistent with the literature.

References:

[1] Dargaville, S. & Farrell, T. W. Predicting Active Material Utilization in LiFePO4 Electrodes Using a Multiscale Mathematical Model. J. Electrochem. Soc. 157, A830 (2010).

[2] Lu, X. et al. Effect of carbon blacks on electrical conduction and conductive binder domain of next-generation lithium-ion batteries. Journal of Power Sources 592, 233916 (2024).

[3] Finley, T. D. (2014). Battery degradation modeling for vehicle applications [Master's thesis, University of Waterloo].